

Course 6: Advanced Features

USP Training Series

Dr. ABDELMALEK OMAR

USP Zephyr Training Series
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Course Overview

- 1 Introduction
- 2 FUOTA (Firmware Update Over-The-Air)
- 3 LoRaWAN Relay
- 4 LoRaWAN Certification
- 5 Production Optimization
- 6 Hands-On Labs
- 7 Summary

Course 6: Advanced Features

Learning Objectives

- Implement Firmware Update Over-The-Air (FUOTA)
- Configure and use LoRaWAN Relay
- Prepare devices for LoRaWAN certification
- Optimize for production deployment
- Handle advanced multicast and class B/C operations

Hands-On Labs

- Lab 6.1: FUOTA Implementation
- Lab 6.2: LoRaWAN Relay Configuration
- Lab 6.3: Certification Testing
- Lab 6.4: Production Optimization

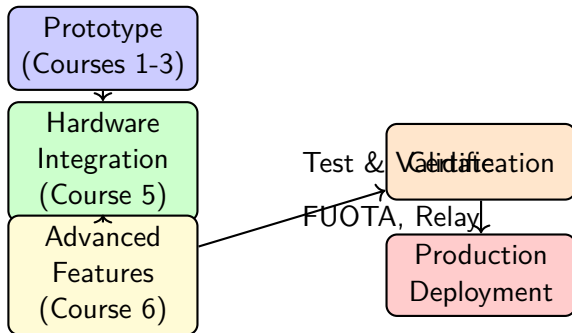
Prerequisites

- Completion of Courses 1-5
- Understanding of LoRaWAN protocol stack
- Experience with LBM and RAC architecture
- Knowledge of device tree and calibration
- Access to LoRaWAN Network Server with FUOTA/Relay support

Advanced Topics

This course covers production-ready features and certification requirements.

Production Deployment Journey



What is FUOTA?

Definition

FUOTA (Firmware Update Over-The-Air) enables remote firmware updates for LoRaWAN devices without physical access.

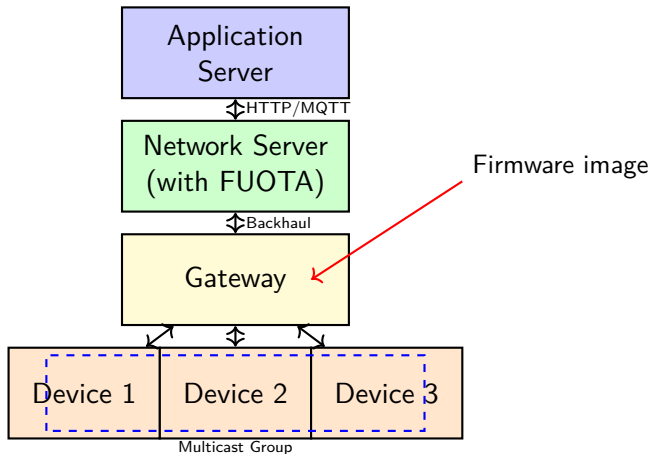
Key Benefits:

- Update deployed devices remotely
- Fix bugs without device retrieval
- Add new features post-deployment
- Reduce maintenance costs
- Extend device lifetime

Use Cases:

- Smart city sensors (thousands of devices)
- Remote agricultural monitoring
- Industrial IoT in hazardous locations
- Underground or underwater deployments

FUOTA Architecture



FUOTA Process Flow

1 Setup Multicast Session

- Network Server creates multicast group
- Devices join multicast session
- Session keys distributed

2 Firmware Fragmentation

- Firmware split into fragments (typ. 200 bytes each)
- Reed-Solomon FEC (Forward Error Correction)
- Redundancy for lost packets

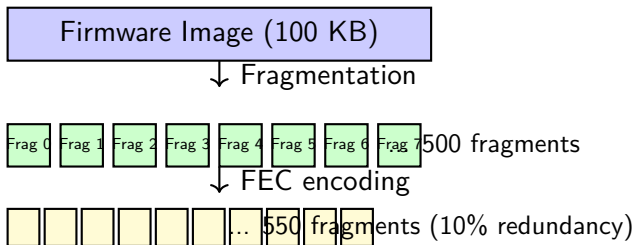
3 Multicast Transmission

- Network Server broadcasts fragments
- Devices collect fragments
- Request missing fragments (unicast)

4 Validation and Installation

- Verify firmware integrity (CRC, signature)
- Install to secondary slot (MCUboot)
- Reboot and swap images

FUOTA Fragmentation



Fragment Size: 200 bytes typical (fits in LoRaWAN payload)

Redundancy: 5-10% extra fragments for error correction

Time Estimate: 100 KB firmware, 500 fragments, 1 per minute = 8 hours

FUOTA Implementation with LBM

```
#include <smtc_modem_api.h>

void fuota_init(void) {
    // Enable multicast
    smtc_modem_multicast_set_grp_config(
        0,    // Stack ID
        0,    // Multicast group index
        MULTICAST_ADDR,
        MULTICAST_NWK_SKEY,
        MULTICAST_APP_SKEY
    );

    // Configure firmware update service
    smtc_modem_file_upload_init(0, CIPHER_MODE_AES_CCM
    );

    // Set firmware size
    smtc_modem_stream_init(0, FW_SIZE,
        STREAM_ENCRYPTION_ENABLE);
```

FUOTA Event Handling

```
void fuota_event_handler(void) {
    smtc_modem_event_t event;

    while (smtc_modem_get_event(&event, &pending)
           == SMTC_MODEM_RC_OK) {
        switch (event.event_type) {
            case SMTC_MODEM_EVENT_STREAM_DONE:
                printk("Firmware download complete!\n");
                verify_and_install_firmware();
                break;

            case SMTC_MODEM_EVENT_UPLOAD_DONE:
                printk("Upload session complete\n");
                break;

            case SMTC_MODEM_EVENT_FIRMWARE_MANAGEMENT:
                handle_firmware_management(&event);
                break;
        }
    }
}
```

MCUboot Integration

```
#include <zephyr/dfu/mcuboot.h>
#include <zephyr/storage/flash_map.h>

void verify_and_install_firmware(void) {
    // Verify firmware signature
    if (!verify_firmware_signature()) {
        printk("Firmware signature invalid!\n");
        return;
    }

    // Mark image as pending (MCUboot will swap)
    int ret = boot_request_upgrade(
        BOOT_UPGRADE_TEST);

    if (ret == 0) {
        printk("Firmware marked for upgrade\n");
        printk("Rebooting in 5 seconds...\n");
        k_sleep(K_SECONDS(5));
        sys_reboot(SYS_REBOOT_WARM);
```

FUOTA Best Practices

Design Considerations

- **Image Size:** Keep firmware < 200 KB to reduce update time
- **Timing:** Schedule updates during low-traffic periods
- **Redundancy:** Use 10% FEC redundancy for reliable transfer
- **Verification:** Always verify signature before installation
- **Rollback:** Implement fallback to previous firmware if update fails

Security

- Use cryptographic signatures (Ed25519 or ECDSA)
- Encrypt firmware images
- Validate firmware before installation
- Protect firmware download from tampering

LoRaWAN Relay Overview

What is LoRaWAN Relay?

LoRaWAN Relay extends network coverage by allowing end devices to relay packets from other devices.

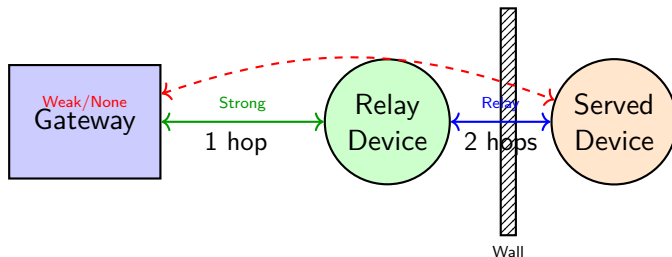
Use Cases:

- Extend coverage in buildings (indoor to outdoor)
- Bridge gaps in network coverage (dead zones)
- Reach devices in basements or tunnels
- Reduce gateway infrastructure costs

Two Modes:

- **Served End Device:** Device with poor coverage (needs relay)
- **Relay Device:** Device with good coverage (provides relay)

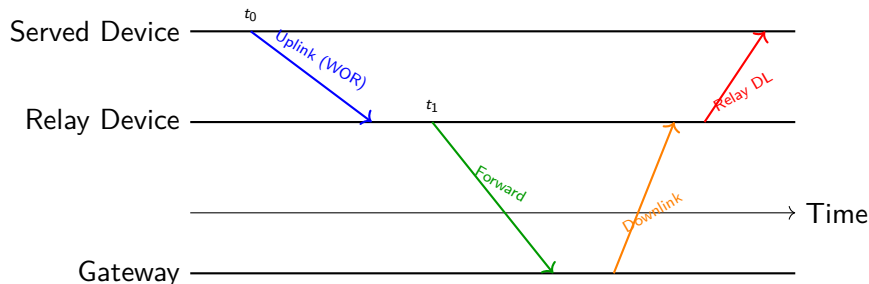
Relay Network Topology



Relay Benefits:

- Served device reaches gateway via relay (2-hop)
- No infrastructure changes needed
- Dynamic relay selection

Relay Packet Flow



Key Concepts:

- **WOR (Wake-On-Radio):** Low-power preamble detection
- **CAD (Channel Activity Detection):** Relay listens for preambles
- **Forwarding:** Relay re-transmits packet to gateway

Configure Relay Mode - Served Device

```
#include <smtc_modem_api.h>
#include <smtc_modem_relay_api.h>

void relay_served_init(void) {
    // Enable relay mode
    smtc_modem_relay_tx_enable(0, true);

    // Configure WOR settings
    smtc_modem_relay_tx_set_activation_mode(
        0, SMTC_MODEM_RELAY_TX_ACTIVATION_MODE_ED);

    // Set relay search period
    smtc_modem_relay_tx_set_smart_level(0, 8);

    // Use WOR channel for uplinks
    smtc_modem_relay_tx_set_backoff(0, 60); // 60s

    printk("Relay served mode enabled\n");
}
```

Configure Relay Mode - Relay Device

```
void relay_device_init(void) {  
    // Enable relay RX mode (forwarding)  
    smtc_modem_relay_rx_enable(0, true);  
  
    // Configure relay behavior  
    smtc_modem_relay_rx_set_activation_mode(  
        0, SMTC_MODEM_RELAY_RX_ACTIVATION_MODE_ENABLE)  
        ;  
  
    // Set forwarding rules  
    smtc_modem_relay_rx_set_smart_level(0, 8);  
  
    // CAD period (wake to listen for WOR)  
    smtc_modem_relay_rx_set_backoff(0, 60); // 60s  
  
    printk("Relay device mode enabled\n");  
}
```

```
void relay_event_handler(void) {
```

Relay Power Consumption

Challenge

Relay devices must wake periodically to detect WOR preambles, increasing power consumption.

Optimization Strategies:

- **Adaptive CAD:** Increase CAD period when no traffic
- **Smart Level:** Adjust relay sensitivity based on RSSI
- **Battery vs. Mains:** Use mains-powered devices as relays
- **Duty Cycling:** Enable relay only during certain hours

Typical Current:

- Normal operation: 5 μ A sleep
- With relay (CAD every 60s): 20 μ A average
- 4x increase in power consumption

Recommendation

Use mains-powered or solar-powered devices as relays whenever possible.

Why LoRaWAN Certification?

Certification Benefits

- Guarantees interoperability with all LoRaWAN networks
- Required by many operators and customers
- Demonstrates compliance with LoRaWAN specification
- Builds trust and credibility
- Access to LoRa Alliance logo and marketing

Certification Levels:

- **Pre-Certification:** Self-testing with open tools
- **Official Certification:** Third-party lab testing
- **Re-Certification:** After major firmware changes

Certification Cost: Typically \$2,000 - \$5,000 USD per device type

Certification Test Categories

1 MAC Layer Compliance

- OTAA join procedure
- Uplink/downlink handling
- Adaptive Data Rate (ADR)
- RX windows timing
- Duty cycle compliance

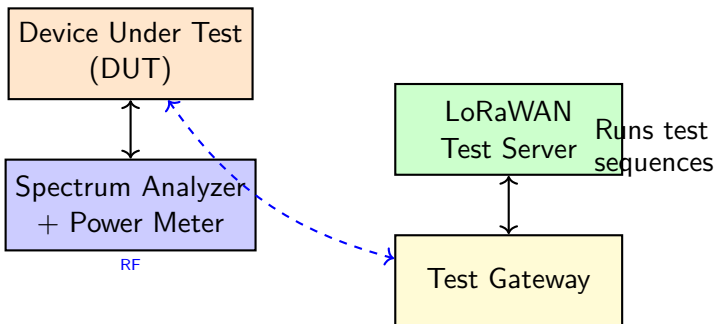
2 PHY Layer Compliance

- Frequency accuracy (± 20 ppm)
- TX power accuracy
- Modulation parameters
- Receive sensitivity

3 Regional Parameters

- EU868, US915, AS923, etc.
- Channel plans
- Duty cycle limits
- Dwell time limits

Certification Testing Setup



Test Tools:

- LoRa Alliance Test Tool (official)
- End Device Certification Test Suite
- RF test equipment (spectrum analyzer, power meter)

Common Certification Failures

1. RX Window Timing

Issue: RX1/RX2 windows not opened at correct time

Fix: Calibrate crystal, improve timing accuracy

2. Duty Cycle Violation

Issue: Exceeds 1% (EU) or dwell time limits

Fix: Implement proper duty cycle tracking

3. TX Power Inaccuracy

Issue: Actual TX power differs from configured

Fix: Calibrate TX power (see Course 5)

4. ADR Not Working

Issue: Device doesn't respond to ADR commands

Fix: Enable ADR in LBM, test with varying RSSI

Prepare for Certification

```
// prj.conf - Enable certification features
```

```
CONFIG_LORAWAN_VERSION_1_0_4=y  
CONFIG_LORAWAN_DUTYCYCLE=y  
CONFIG_LORAWAN_ADR=y  
CONFIG_LORAWAN_PUBLIC_NETWORK=y
```

```
// Enable all supported regions
```

```
CONFIG_LORAWAN_REGION_EU868=y  
CONFIG_LORAWAN_REGION_US915=y  
CONFIG_LORAWAN_REGION_AS923=y
```

```
// Enable class B/C if supported
```

```
CONFIG_LORAWAN_CLASS_B=y  
CONFIG_LORAWAN_CLASS_C=y
```

```
// Enable proper NVM for persistent storage
```

```
CONFIG_NVS=y  
CONFIG_LORAWAN_NVM_CONTEXT_MGMT=y
```

Production Checklist

1 Security

- Unique DevEUI, JoinEUI, AppKey per device
- Secure key storage (e.g., secure element)
- Encrypted firmware images
- Disable debug interfaces

2 Power Optimization

- Enable low-power modes (Course 2)
- Optimize uplink interval
- Use confirmed uplinks sparingly
- Implement adaptive strategies

3 Reliability

- Implement watchdog timer
- Handle join failures gracefully
- Store critical data in NVM
- Test in real-world conditions

CRITICAL: Unique Keys

Every device **MUST** have unique:

- DevEUI (globally unique 64-bit ID)
- JoinEUI (formerly AppEUI)
- AppKey (128-bit secret key)

Never reuse keys across devices!

Key Provisioning Options:

- 1 **Pre-Provisioned:** Keys programmed during manufacturing
- 2 **Secure Element:** Keys stored in ATECC608 or similar
- 3 **On-Demand:** Keys generated and assigned during commissioning

Protect Keys:

- Store in flash with read-out protection
- Use MPU/TrustZone if available
- Disable JTAG/SWD in production

Provision Unique Keys

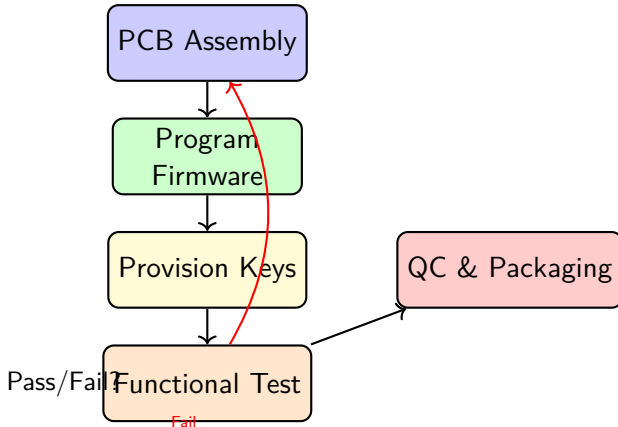
```
// Read DevEUI from secure storage or MCU unique ID
void get_device_eui(uint8_t *dev_eui) {
    // Option 1: Use MCU unique ID
    uint32_t *uid = (uint32_t *)0xFE081F0; //
        nRF54L15
    memcpy(dev_eui, uid, 8);

    // Option 2: Read from secure element
    // atecc608_read_serial(dev_eui);

    // Option 3: Read from provisioned flash region
    // flash_read(PROVISIONING_ADDR, dev_eui, 8);
}

void lorawan_provision(void) {
    uint8_t dev_eui[8];
    uint8_t join_eui[8] = LORAWAN_JOIN_EUI;
    uint8_t app_key[16] = LORAWAN_APP_KEY;
}
```

Manufacturing Test Flow



Functional Tests:

- Read chip ID (verify communication)
- Perform LoRaWAN join
- Send test unlink

Field Deployment Considerations

Environmental:

- IP67/IP68 enclosure
- Operating temp range
- UV resistance
- Moisture protection
- Vibration tolerance

Antenna:

- External vs. internal
- Vertical polarization
- Ground plane requirements
- Avoid metal enclosures

Battery:

- Li-SOCl₂ for long life
- LiPo for rechargeable
- Capacity calculation
- Temperature derating

Installation:

- Outdoor mounting
- Line-of-sight to gateway
- Avoid Faraday cages
- Test coverage before install
- Document GPS coordinates

Lab 6.1: FUOTA Implementation

Objective

Implement firmware update over-the-air using multicast and LBM.

Tasks:

- 1 Enable MCUboot bootloader with dual-image support
- 2 Configure LBM for multicast and file upload
- 3 Join LoRaWAN network
- 4 Configure network server to send firmware update
- 5 Receive firmware fragments via multicast
- 6 Verify and install updated firmware
- 7 Test rollback if update fails

Deliverables:

- FUOTA-enabled firmware
- Successful OTA update demonstration
- Serial logs showing update process

Lab 6.2: LoRaWAN Relay Configuration

Objective

Configure one device as relay and another as served device to extend coverage.

Tasks:

- 1 Configure Device A as relay device (good coverage location)
- 2 Configure Device B as served device (poor coverage location)
- 3 Test direct uplink from Device B (expect failure/weak signal)
- 4 Enable relay mode on both devices
- 5 Test uplink from Device B via relay (expect success)
- 6 Monitor relay events and packet forwarding
- 7 Measure additional power consumption on relay device

Deliverables:

- Relay and served device firmware
- Coverage comparison (with/without relay)
- Power consumption analysis

Lab 6.3: Certification Pre-Testing

Objective

Perform self-testing to prepare for official LoRaWAN certification.

Tasks:

- 1 Download LoRa Alliance End Device Test Tool
- 2 Set up local LoRaWAN test network
- 3 Run MAC compliance tests (OTAA, ADR, duty cycle)
- 4 Run PHY compliance tests (frequency, power, sensitivity)
- 5 Run regional parameter tests (EU868 or US915)
- 6 Document any failures and fix issues
- 7 Re-test until all tests pass

Deliverables:

- Test results report
- List of issues found and fixes applied
- Final passing test results

Lab 6.4: Production Optimization

Objective

Optimize firmware for production deployment with security and power savings.

Tasks:

- 1 Implement unique key provisioning from MCU UID
- 2 Enable secure boot and flash read-out protection
- 3 Optimize power consumption (target $< 10 \mu\text{A}$ average)
- 4 Add watchdog timer and error recovery
- 5 Implement persistent storage for critical state
- 6 Create manufacturing test firmware
- 7 Measure battery life estimate (with profiler)

Deliverables:

- Production-ready firmware
- Power consumption report
- Battery life calculation

Course 6 Summary

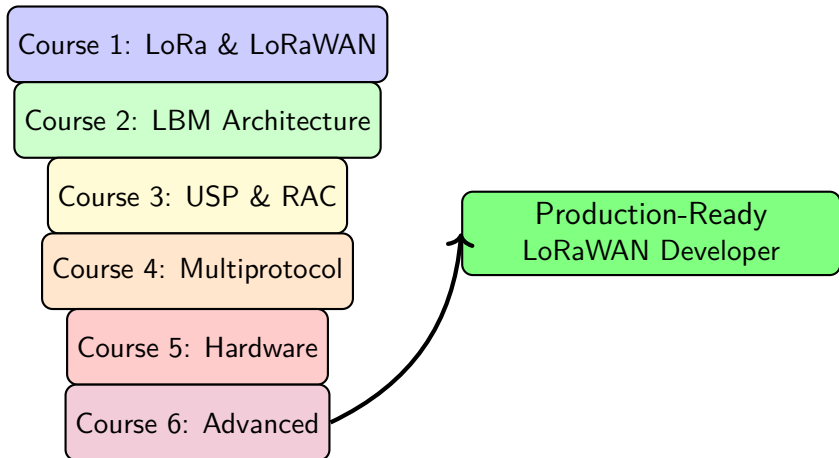
Key Takeaways

- FUOTA enables remote firmware updates for deployed devices
- LoRaWAN Relay extends coverage without additional infrastructure
- Certification ensures interoperability and compliance
- Production requires unique keys, security, and optimization
- Real-world deployment needs environmental and RF considerations

Congratulations!

- You've completed all 6 USP training courses
- You're ready to develop production LoRaWAN devices
- Continue learning with advanced topics and certifications

Training Series Complete



Additional Resources

Documentation:

- LoRa Alliance Specifications:
- LBM Documentation:
- Zephyr RTOS Docs:
- MCUboot:

Certification:

- LoRa Alliance Device Certification:
- End Device Test Tool

Community:

- LoRa Alliance Technical Committee
- The Things Network Community
- Zephyr Discord

Thank You!

Congratulations on Completing
USP Training Series

Questions or Support:

`usp-support@example.com`

Lab Solutions:

`doc/training/Lab_Solutions.md`